SAVERIO MONACO

Curriculum vitae et studiorum

PERSONAL INFORMATION

Via Pozzo 64, San Giovanni La Punta, Catania

+39 3922660790

saveriomonaco97@gmail.com

https://github.com/SaverioMonaco

https://www.linkedin.com/in/saverio-monaco/

Date of birth 15 August 1997 | Nationality Italian

EDUCATION AND TRAINING

2020–Today

Master in Physics of Data

Università degli Studi di Padova

Padua – Padova, Italy

2020–Today

University Merit College "Don Nicola Mazza"

Padua - Padova, Italy

2016–2020

Bachelor of Physics

Università degli Studi di Catania

Thesis: Phase-Space Formulation of Quantum Mechanics.
A concise guide to an alternative and equivalent formulation of Quantum Mechanics in which position and momentum variables are expressed through Wigner quasiprobability distributions.

Catania, Italy

2012–2016

Convitto Cutelli High School

Graduated with 100/100

Received French double degree ESABAC

2014–2015

Exchange year in Germany

High School: Gymnasium der Stadt Rahden, Germany
Fourth year of High School attended in Stemshorn, Germany

PERSONAL SKILLS

Mother tongue: Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	C1	C1	C1
	IELTS Academic: score 7				
German	C1	B2	C1	B2	B2
	Goethe-Zertifikat B2				
French	B2	B2	B1	B1	B2
	EsaBac diploma				

Computer skills

- Excellent knowledge of the language Python (including the main libraries such as Pandas, Numpy, Plotly and Seaborn)
- Excellent knowledge of Machine Learning and Deep Learning libraries: scikit-learn and PyTorch
- Good knowledge of C, C++, R, VHDL, Fortran.
- Excellent knowledge of LaTeX
- Good knowledge of Git
- Good knowledge of Linux
- ECDL certification.

Driving licence B

MAIN PROJECTS

2022 Random Forest Regressor for SEVN library

Implementation in Python of a Machine Learning Regressor model for predicting the convective parameters of stars from SEVN population-syntesis code.
A paper is being written about this work.

2021 Implementation of two Convolutional Neural Networks for MAGIC Telescopes

Implementation in Python and CTLearn of two distinct Convolutional Neural Networks for Gamma-ray/Cosmic-ray classifier and direction-energy reconstruction using simulated and real data from MAGIC Cherenkov Telescopes.

2021 4-Taps Finite Impulse Response Filter in VHDL

Implementation of a UART and Low-Pass filter in a FPGA using VHDL.

2021 Processing of streaming data using Apache Spark and Kafka

Final group-project for the analysis through Parallel Computing of simulated streaming processing of cosmic rays in Drift Tubes detectors. Implementation of a live dashboard using Plotly Dash.

2021 Image Colorization through Segmentation

Realization of a Self-Supervised Convolutional Neural Network in PyTorch for coloring grayscale images and videos using a Segmentation Network on top for better performances.

2020 FlowSolver

Implementation in C of a fully-optimized recursive algorithm for solving *Flow Free* type of puzzles, and creation of a terminal version of the game.